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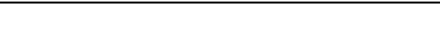
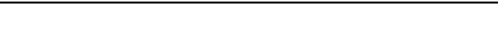
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Client: MMA
Project: mSafe2
Description Document: 75287 DIAGRAMA ELÉCTRICO mSafe2

Document Number: 75287 ZPL
Revision: 6
Date: 26-04-2026

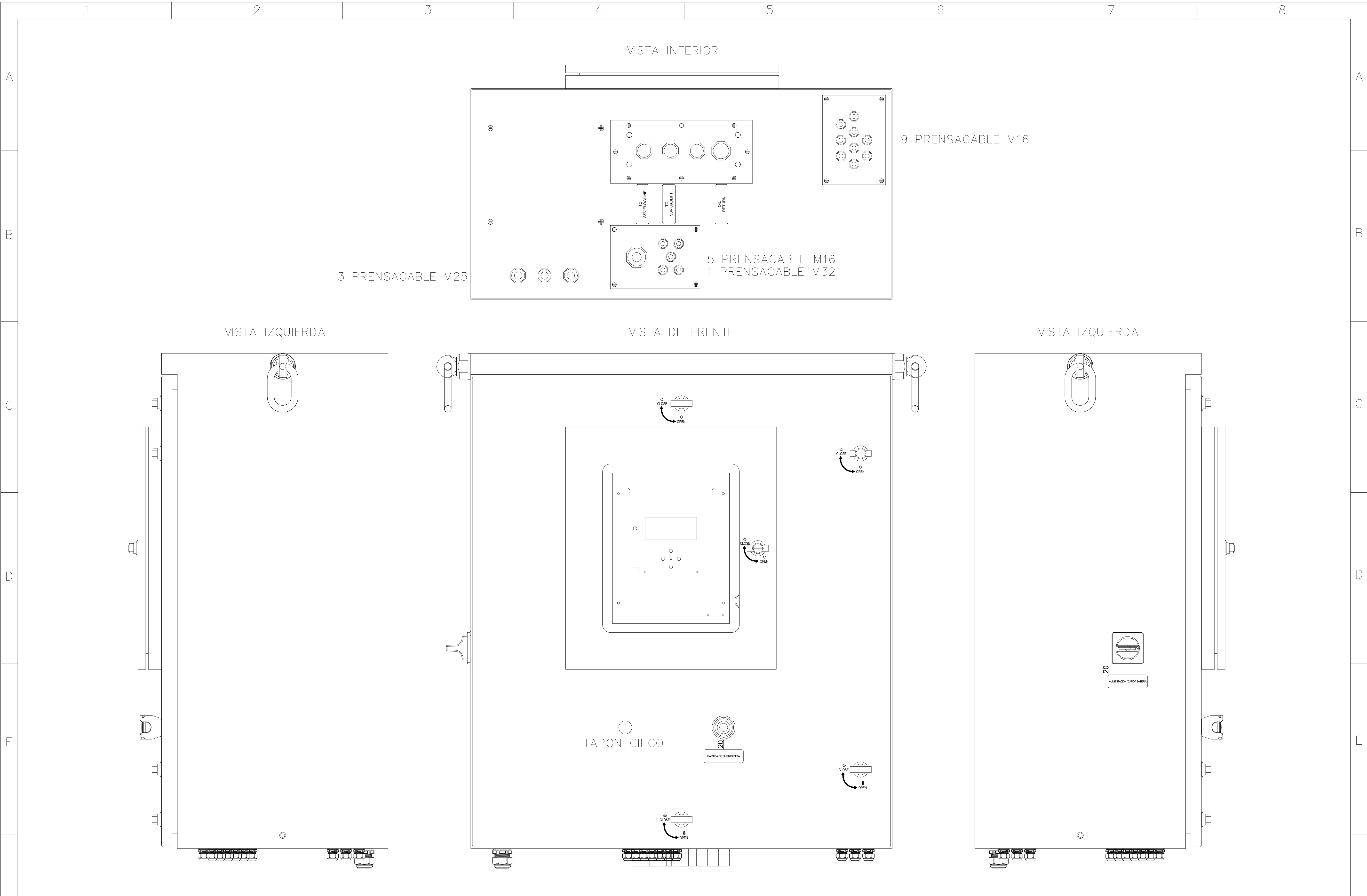
6	Update whit fixes	13/07/26	LM	MD	MPV	PROJECT: C.J. CONTROLADOR mSafe2
5	Update whit fixes	26/05/26	LM	MD	MPV	CODE: 75286-1
4	Update whit fixes	15/05/26	LM	MD	MPV	CLIENT:
3	For Manufacturing	12/03/26	LM	MD	MPV	MMA
REV	DESCRIPTION	DATE	PROJ.	DW.	APR.	

 BATFER Investment S.A.		DOC NUMBER:		75287		GRAL REV: 6	
		TITLE:		DIAGRAMA ELÉCTRICO mSafe2		PAG:	REV:
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				—		CONT:	01

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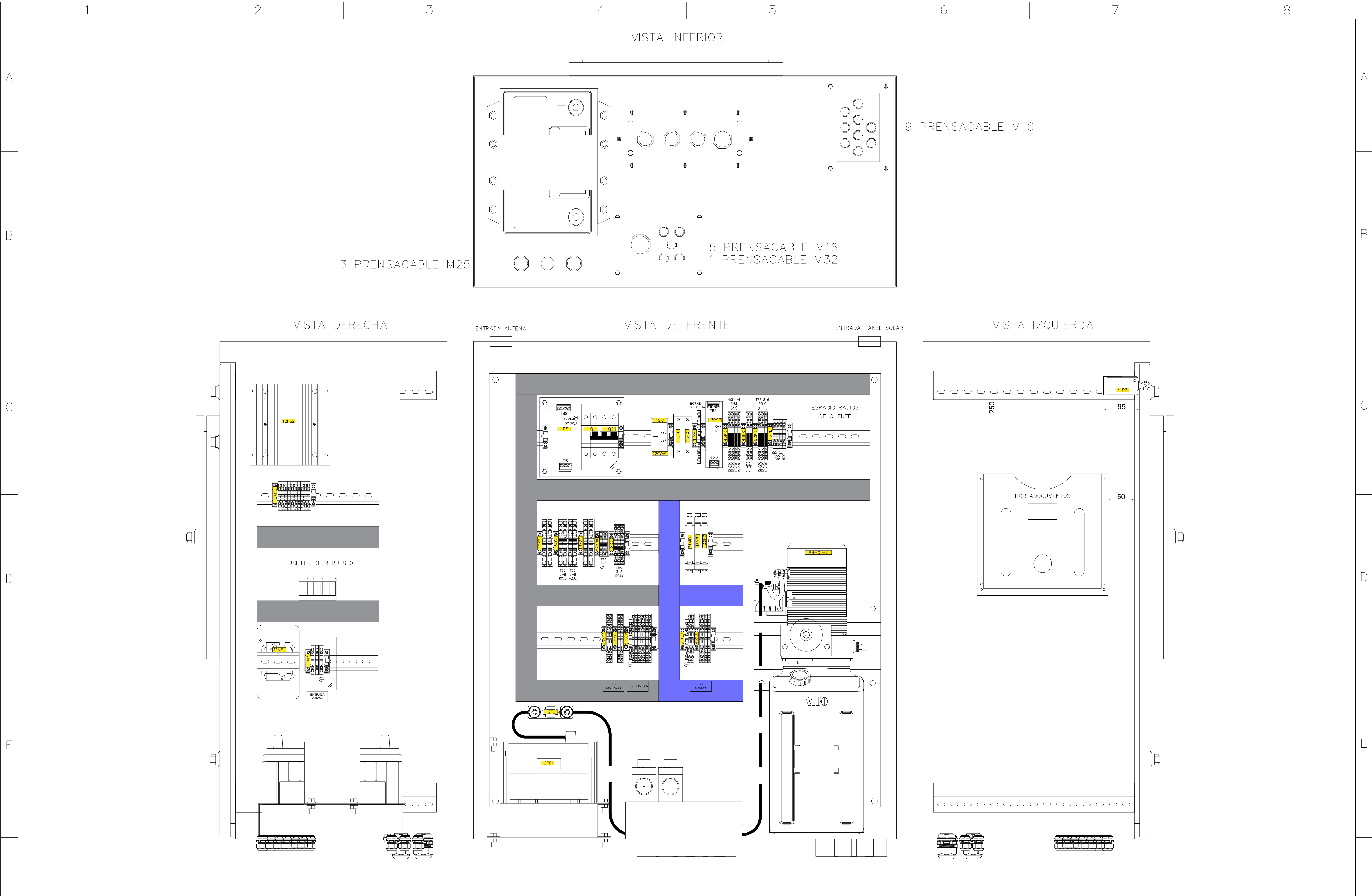
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3	For Manufacturing	12/03/26	LM	MD	MPV
REV	DESCRIPTION	DATE	PROJ.	DW.	APR.

PROJECT: C.J. CONTROLADOR mSafe2
CODE: 75286-1
CLIENT: MMA



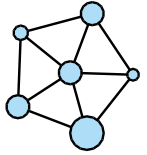
DOC NUMBER:	75287	GRAL REV:	6
TITLE:	DIAGRAMA ELÉCTRICO mSafe2 TOPOGRÁFICO EXTERIOR	PAG:	02
		REV:	6
		CONT:	03

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6	Update whit fixes	13/07/26	LM	MD	MPV
5	Update whit fixes	26/05/26	LM	MD	MPV
4	Update whit fixes	15/05/26	LM	MD	MPV
3	For Manufacturing	12/03/26	LM	MD	MPV
REV	DESCRIPTION	DATE	PROJ.	DW.	APR.

PROJECT: C.J. CONTROLADOR mSafe2
CODE: 75286-1
CLIENT: MMA



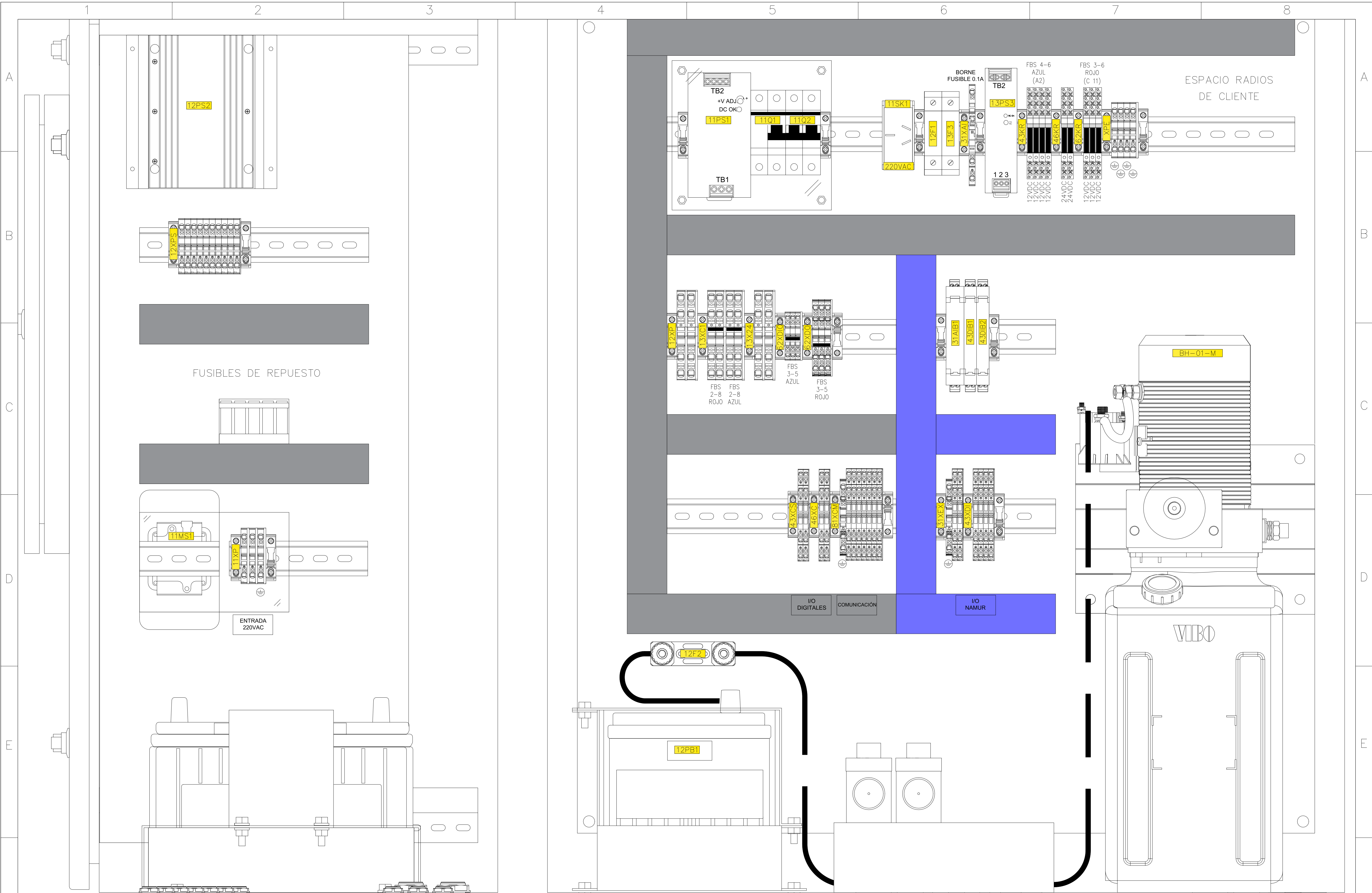
BATFER
Investment S.A.



DOC NUMBER:	75287	GRAL REV:	6
TITLE:	DIAGRAMA ELÉCTRICO mSafe2 TOPOGRÁFICO INTERIOR	PAG:	03
		REV:	6
		CONT:	04

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NOTE: cables not indicated will be black 0.75mm²/18 AWG



6	Update whit fixes	13/07/26	LM	MD	MPV
5	Update whit fixes	26/05/26	LM	MD	MPV
4	Update whit fixes	15/05/26	LM	MD	MPV
3	For Manufacturing	12/03/26	LM	MD	MPV
REV	DESCRIPTION	DATE	PROJ.	DW.	APR.

PROJECT: C.J. CONTROLADOR mSafe2
CODE: 75286-1
CLIENT: MMA



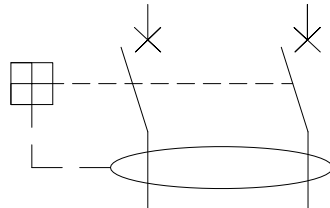
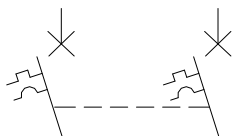
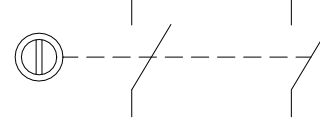
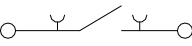
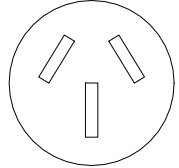
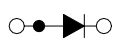
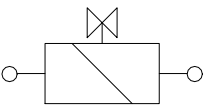
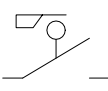
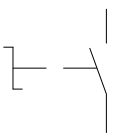
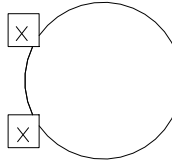
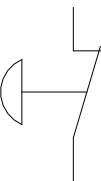
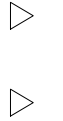
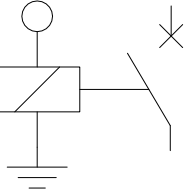
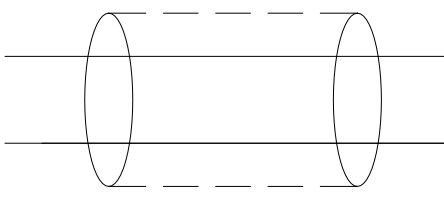
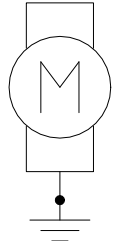
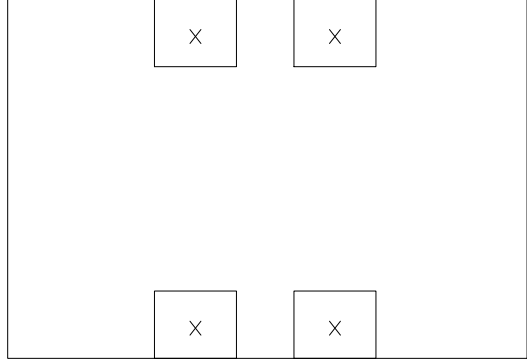
DOC NUMBER: 75287
TITLE: DIAGRAMA ELÉCTRICO mSafe2
TOPOGRÁFICO INTERIOR
DETALLES

GRAL REV:	6
PAG:	04
REV:	6
CONT:	09

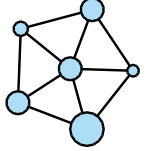
NOTE: cables not indicated will be black 0.75mm ² /18 AWG															
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A	TAG	DESCRIPTION					BRAND			MODEL / CODE					A
	11XP	Bornes de alimentación					PHOENIX CONTACT			PT 6					
	11MS1	Interruptor Seccionador					SCHNEIDER			VBF1					
	11PS1	Fuente de alimentación 110–240 Vac a 24 VDC					MEAN WELL			NDR–120–24					
	11Q1	Disyuntor diferencial 25A – 30mA					SCHNEIDER			EZ9R36225					
	11Q2	Interruptor termomagnético 2x10A curva C					SCHNEIDER			EZ9F34210					
	11SK1	Toma corriente IRAM					VIVION			–					
B	BH–01–ZV	Contactor de bomba de 12 VDC					TROMBETTA			684–1261–212–17					B
	12XPS	Bornes de cargador de Batería					PHOENIX CONTACT			PT 6					
	12PS2	Cargador de Baterías					EPEVER			MPPT TRACER5210BP					
	12XP	Bornes de distribución de voltaje de 12VDC					PHOENIX CONTACT			PT 6 QUATTRO					
	12F1	Base de fusible para alimentación de batería					SCHNEIDER			DF101 10x38					
	–	Fusible cerámico cilíndrico 20A 10x38mm gL					–			–					
	12F2	Base para fusible tipo MEGA					MEGA			–					
	–	Fusible tipo MEGA 150A					–			–					
	13XC1	Distribución de voltaje de control 12VDC					PHOENIX CONTACT			PT 6 QUATTRO					
	13PS2	Conversor 12/24VDC					MEAN WELL			DDR–120A–24					
C	13F3	Base de fusible para alimentacion 12VDC					SCHNEIDER			DF101 10x38					C
	–	Fusible cerámico cilíndrico 10A 10x38mm gL					–			–					
	13F4	Base de fusible para alimentacion 12VDC					SCHNEIDER			DF101 10x38					
	–	Fusible cerámico cilíndrico 10A 10x38mm gL					–			–					
	13X24	Distribución de voltaje de control 24VDC					PHOENIX CONTACT			PT 6 QUATTRO					
	21PCB01	Placa Electrónica de Control Principal					MMA			L03–60591–01					
	31XAI	Bornes de entradas analógicas internas					PHOENIX CONTACT			PTT 2,5–2MT					
	PT–001	Transmisor de presión de la bomba hidráulica					NOSHOK			PT20–250bar–1–1–10–8					
	31XAI	Bornes de entradas analógicas externas					PHOENIX CONTACT			PTT 2,5–2 MT					
	31XEX	Bornes de entradas analógicas externas					PHOENIX CONTACT			PTT 2,5–2 MT BU					
D	31AIB1	Entrada Analógica – Barrera Seg. Intrínseca					CHENZHU			GS8536–EX					D
	41DS	Interruptor límite de puerta					OMRON			HL–5200					
	43XDI	Bornes de entradas digitales					PHOENIX CONTACT			PTT 2,5–2 MT BU					
	43DIB1/2	Entrada Digital – Barrera Seg. Intrínseca					CHENZHU			GS8512–EX.22					
	43KR	Borne relé 12VDC					PHOENIX CONTACT			RIF–0–RPT–12DC/21					
	46DB1	Botón de paro de emergencia.					SCHNEIDER			ZBE–101					
	46XC	Bornes de comando externo 24 VDC					PHOENIX CONTACT			PTT 2,5–2 MT					
	46KR	Borne relé 24VDC					PHOENIX CONTACT			RIF–0–RPT–24DC/21					
	62KR	Borne relé 12VDC					PHOENIX CONTACT			RIF–0–RPT–12DC/21					
	62XDO	Bornes de comandos					PHOENIX CONTACT			PTT 2,5					
E	62XDIO	Bornes diodo antiparalelo de supresion					PHOENIX CONTACT			PT 2,5–DIO/L–R					E
	SP–1/2	Válvula solenoide					–			–					
	81XCM	Borne de comunicación					PHOENIX CONTACT			PTT 2,5–2MT					
	BORNES DE TIERRA	Borne doble de tierra					PHOENIX CONTACT			PT 2,5–QUATTRO–PE					
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5	Update whit fixes	26/05/26	LM	MD	MPV	CODE: 75286–1			TITLE: DIAGRAMA ELÉCTRICO mSafe2			PAG: 09	REV: 6		
4	Update whit fixes	15/05/26	LM	MD	MPV	CLIENT: MMA			Lista de materiales.						
3	For Manufacturing	12/03/26	LM	MD	MPV										
REV	DESCRIPTION	DATE	PROJ.	DW.	APR.							CONT: 10			
BATFER Investment S.A. DIGITO										75287				6	

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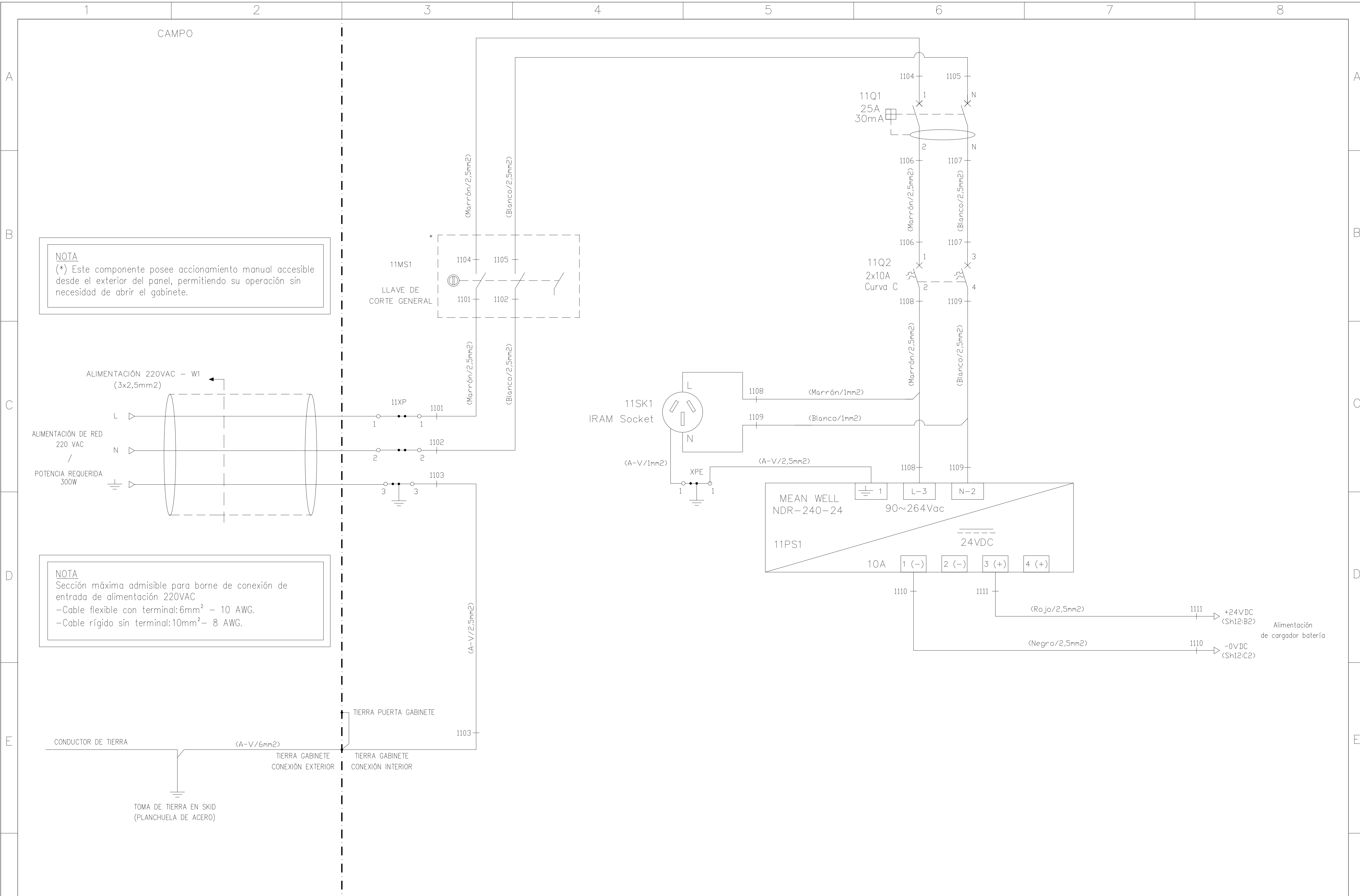
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1	2	3	4	5	6	7	8
A	Símbolo	Descripción	Símbolo	Descripción	Símbolo	Descripción	
B		Diferencial / Disyuntor		Borne Doble		Tag de cable	
		Termomagnética		Borne simple			X X X X
		Seccionador Rotativo		Borne Seccionable		N° de página N° correlativo	
C		Tomacorriente		Borne con diodo			
		Fusible seccionable		Electroválvula		Resistencia	
		Fusible fijo		Switch		Tierra	
D		Selector		Instrumento			
		Parada de Emergencia (Golpe de puño tipo Hongo)		Referencias			
		Bobina de Motor		Cable mallado			
E		Motor Bomba Hidráulica		Componente Simbolo general			

6	Update whit fixes	13/07/26	LM	MD	MPV	PROJECT: C.J. CONTROLADOR mSafe2
5	Update whit fixes	26/05/26	LM	MD	MPV	CODE: 75286-1
4	Update whit fixes	15/05/26	LM	MD	MPV	CLIENT:
3	For Manufacturing	12/03/26	LM	MD	MPV	MMA
REV	DESCRIPTION	DATE	PROJ.	DW.	APR.	

 BATFER Investment S.A.		DOC NUMBER: 75287	GRAL REV: 6
		TITLE: DIAGRAMA ELÉCTRICO mSafe2 Referencia de Símbolos	PAG: 10 REV: 6
			CONT: 11

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5	Update whit fixes	26/05/26	LM	MD	MPV	CODE: 75286-1
4	Update whit fixes	15/05/26	LM	MD	MPV	CLIENT:
3	For Manufacturing	12/03/26	LM	MD	MPV	
REV	DESCRIPTION	DATE	PROJ.	DW.	APR.	MMA



DOC NUMBER: 75287

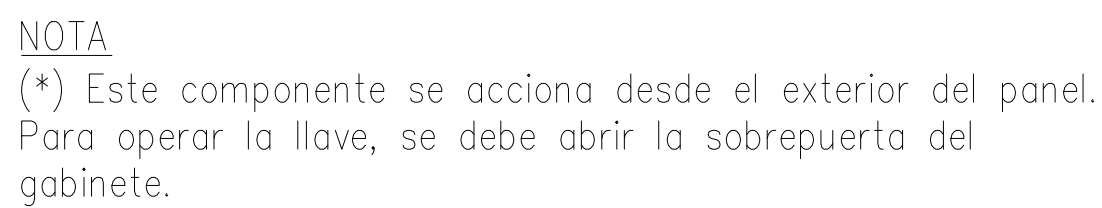
TITLE: DIAGRAMA ELÉCTRICO mSafe2
Alimentación y distribución 220 VAC.

GRAL REV: 6

PAG: 11

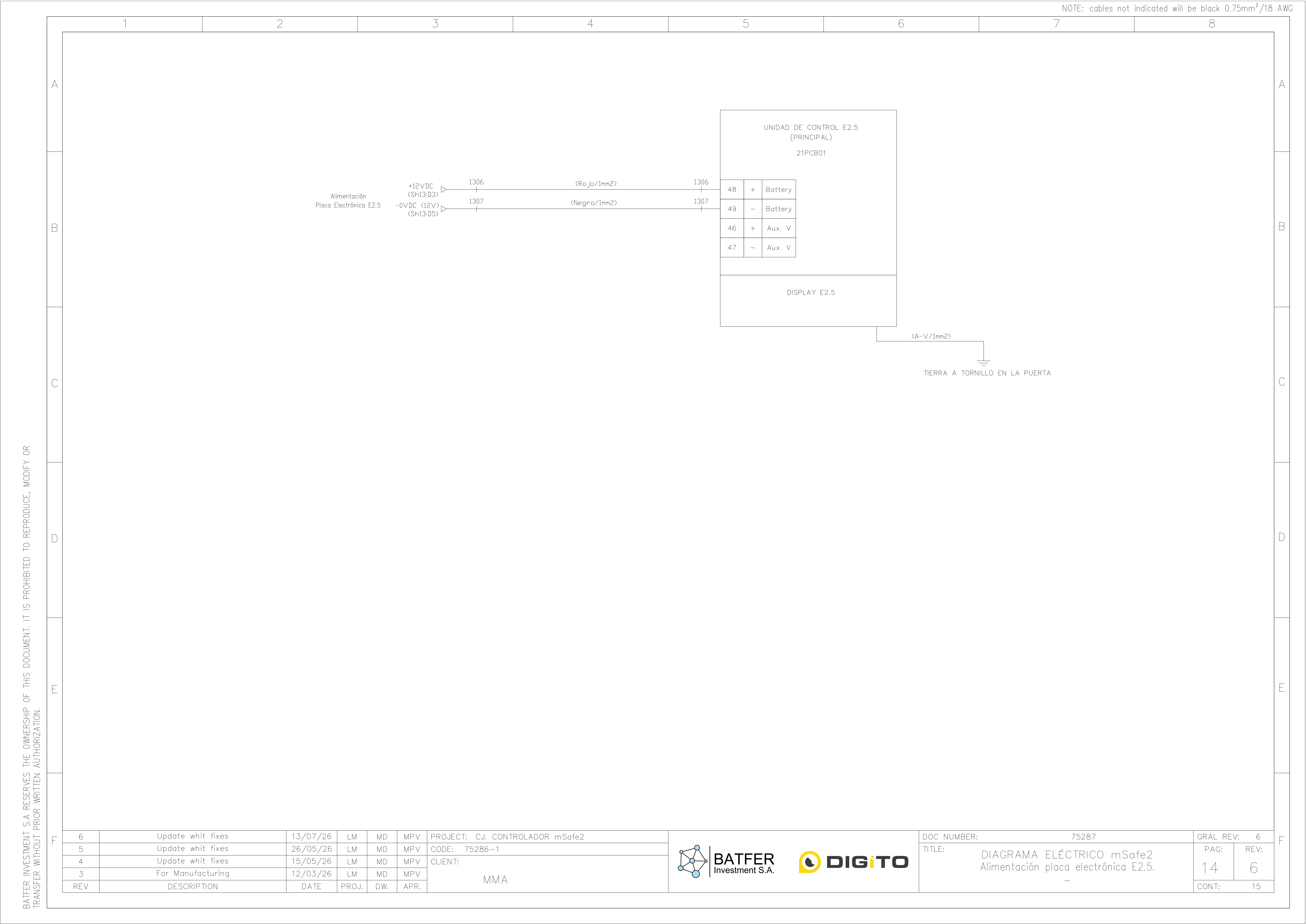
REV: 6

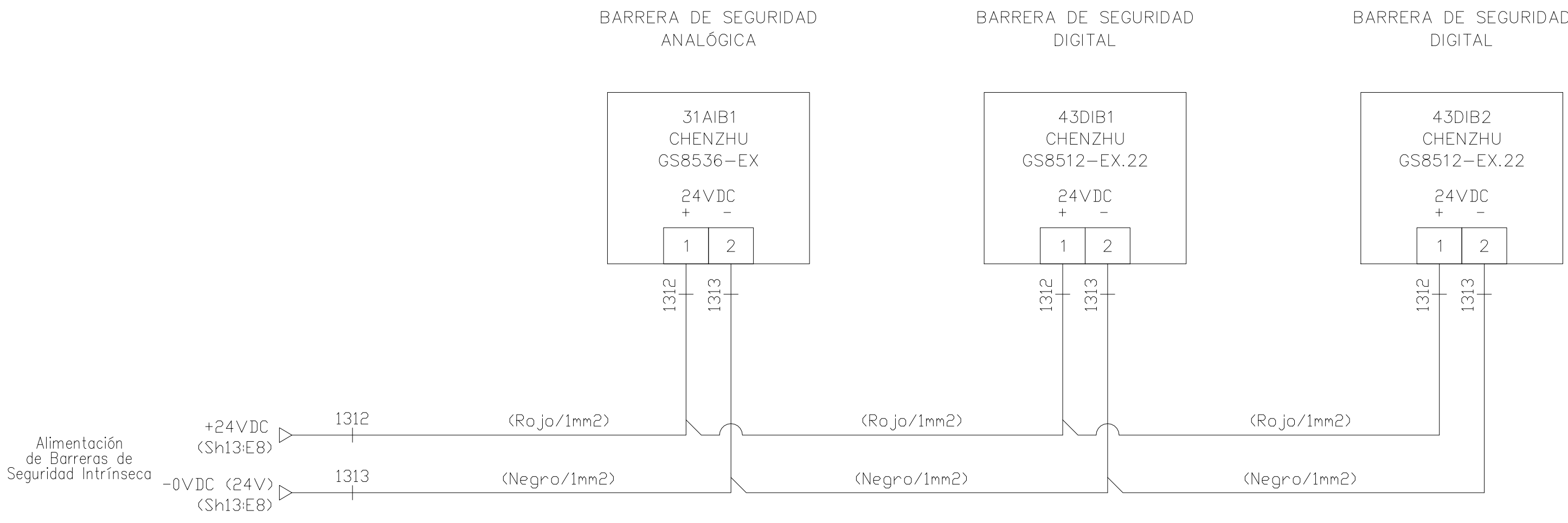
CONT: 12



The footer contains the logos of the two entities. On the left is the logo for BATFER Investment S.A., which consists of a stylized network of blue nodes connected by lines, followed by the text "BATFER" in a large, bold, sans-serif font and "Investment S.A." in a smaller font below it. On the right is the logo for DIGITO, featuring a yellow circular icon with a black shape inside, followed by the word "DIGITO" in a bold, sans-serif font.

DOC NUMBER:	75287	GRAL REV:	6
TITLE:	DIAGRAMA ELÉCTRICO mSafe2 Distribución de voltaje de control.	PAG:	REV:
	—	13	6
		CONT:	14





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6	Update whit fixes	13/07/26	LM	MD	MPV	PROJECT: C.J. CONTROLADOR mSafe2
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3	For Manufacturing	12/03/26	LM	MD	MPV	
REV	DESCRIPTION	DATE	PROJ.	DW.	APR.	



DOC NUMBER: 75287

TITLE: DIAGRAMA ELÉCTRICO mSafe2
Alimentación de barreras de seguridad intrínseca.

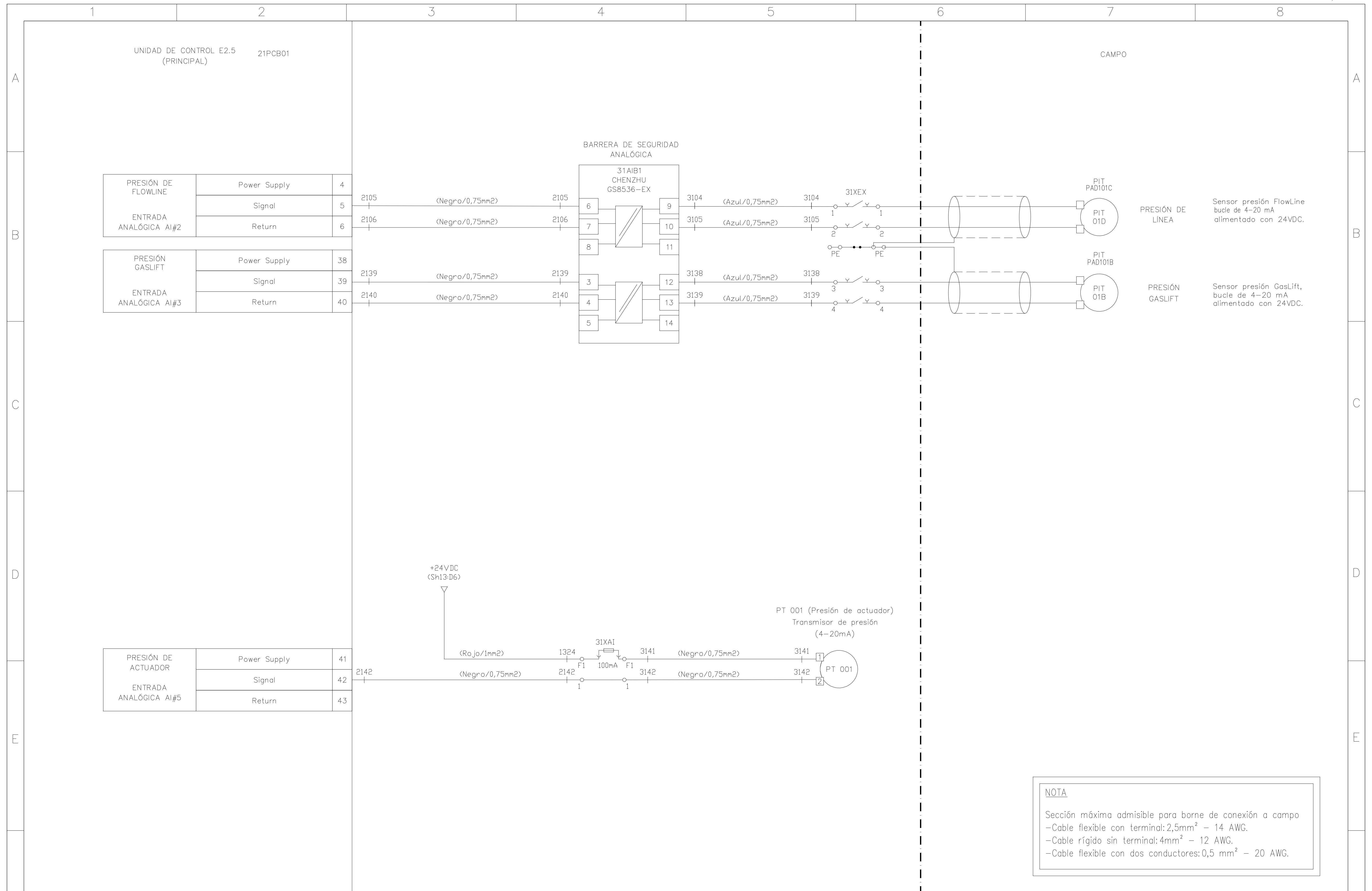
GRAL REV: 6

PAG: 15

REV: 6

CONT: 21

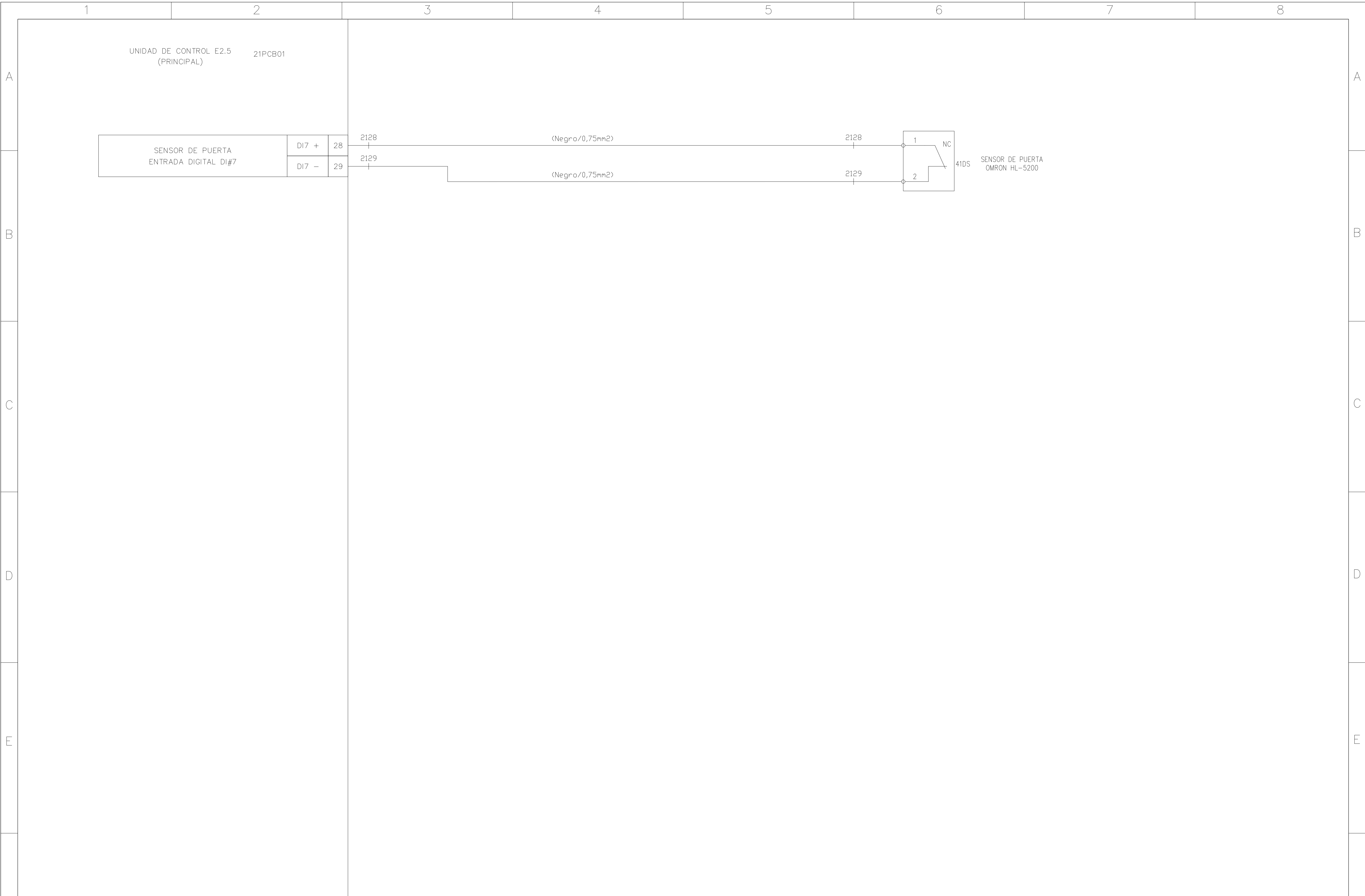




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5	Update whit fixes	26/05/26	LM	MD	MPV	CODE: 75286-1
4	Update whit fixes	15/05/26	LM	MD	MPV	CLIENT:
3	For Manufacturing	12/03/26	LM	MD	MPV	MMA
REV	DESCRIPTION	DATE	PROJ.	DW.	APR.	



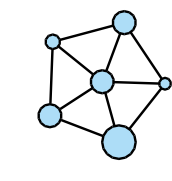
DOC NUMBER:	75287	GRAL REV: 6	
TITLE:	DIAGRAMA ELÉCTRICO mSafe2 Entradas analógicas internas Placa electrónica E2.5.	PAG: 31	REV: 6
		CONT:	33



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6	Update whit fixes	13/07/26	LM	MD	MPV
5	Update whit fixes	26/05/26	LM	MD	MPV
4	Update whit fixes	15/05/26	LM	MD	MPV
3	For Manufacturing	12/03/26	LM	MD	MPV
REV	DESCRIPTION	DATE	PROJ.	DW.	APR.

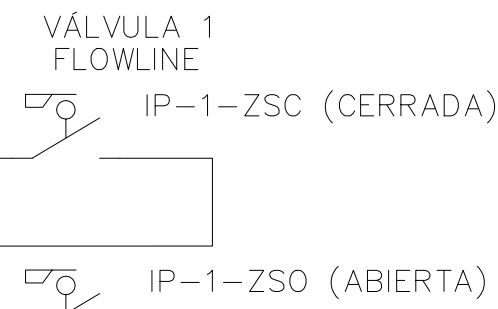
PROJECT: C.J. CONTROLADOR mSafe2
CODE: 75286-1
CLIENT: MMA



BATFER
Investment S.A.



DOC NUMBER: 75287	GRAL REV: 6
TITLE: DIAGRAMA ELÉCTRICO mSafe2 Entradas digitales internas Placa electrónica E2.5.	PAG: 41 REV: 6
CONT: 43	

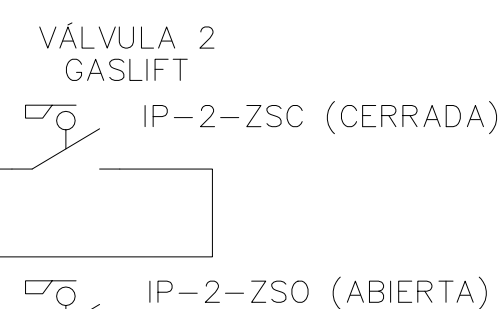


Sensor de posición abierta válvula 1
Interruptor tipo Reed Switch (NA),
protegido con seguridad intrínseca "Ex ia".
Señal alimentada con 24VDC.

NOTA

Sección máxima admisible para borne de conexión a campo

- Cable flexible con terminal: $2,5\text{mm}^2$ - 14 AWG.
- Cable rígido sin terminal: 4mm^2 - 12 AWG.
- Cable flexible con dos conductores: $0,5\text{mm}^2$ - 20 AWG.



Sensor de posición abierta válvula 2
Interruptor tipo Reed Switch (NA),
protegido con seguridad intrínseca "Ex ia".
Señal alimentada con 24VDC.

Feedback de posición válvula abierta
Contacto seco NA
GASLIFT



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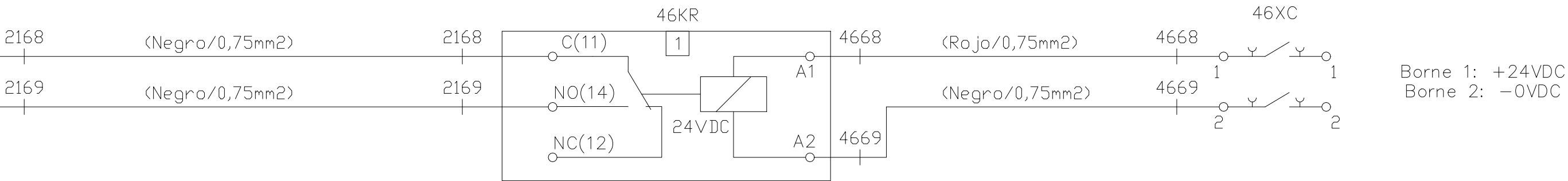
NOTE: cables not indicated will be black 0.75mm²/18 AWG

UNIDAD DE CONTROL E2.5
(PRINCIPAL) 21PCB01

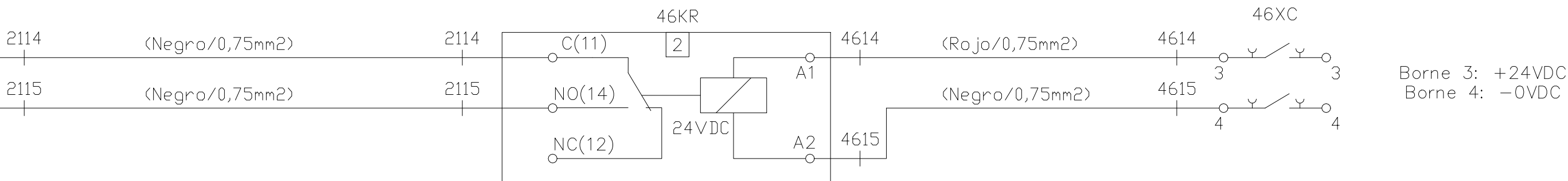
PULSADOR DE SHUTDOWN ENTRADA DIGITAL DI#8	DI8 +	30
	DI8 -	31



COMANDO PERMISIVO EXTERNO 1 ENTRADA DIGITAL DI#10	DI10 +	68
	DI10 -	69



COMANDO PERMISIVO GENERAL ENTRADA DIGITAL DI#11	DI11 +	14
	DI11 -	15



SIN FUNCIÓN
RESERVA

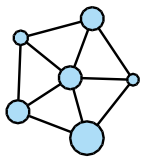
COMANDO PERMISIVO GENERAL
SEÑAL ALIMENTADA CON 24VDC

NOTA
Sección máxima admisible
para borne de conexión a campo
-Cable flexible con terminal: 2,5 mm²
-Cable rígido sin terminal: 4 mm²
-Cable flexible con dos conductores: 0,5 mm²

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6	Update whit fixes	13/07/26	LM	MD	MPV
5	Update whit fixes	26/05/26	LM	MD	MPV
4	Update whit fixes	15/05/26	LM	MD	MPV
3	For Manufacturing	12/03/26	LM	MD	MPV
REV	DESCRIPTION	DATE	PROJ.	DW.	APR.

PROJECT: C.J. CONTROLADOR mSafe2
CODE: 75286-1
CLIENT: MMA

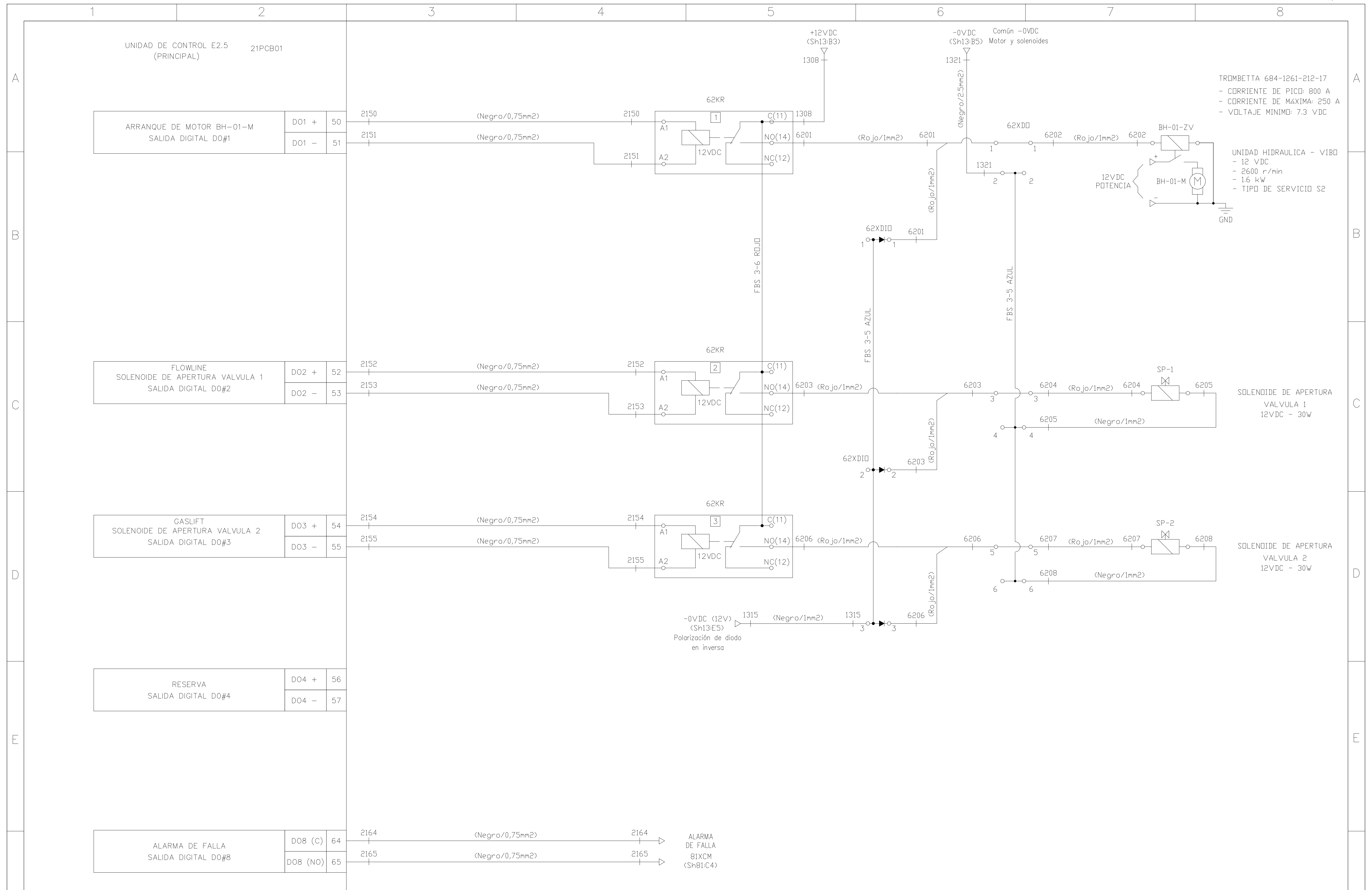


BATFER
Investment S.A.



DOC NUMBER: 75287	GRAL REV: 6
TITLE: DIAGRAMA ELÉCTRICO mSafe2 Entradas digitales externas.	PAG: 46 REV: 6
	CONT: 62

NOTE: cables not indicated will be black 0.75mm²/18 AWG



6	Update whit fixes	13/07/26	LM	MD	MPV	PROJECT: C.J. CONTROLADOR mSafe2	 BATFER Investment S.A.  DIGITO	DOC NUMBER: 75287		GRAL REV: 6	
5	Update whit fixes	26/05/26	LM	MD	MPV	CODE: 75286-1		TITLE: DIAGRAMA ELÉCTRICO mSafe2 Salidas digitales internas. -	PAG:	REV:	
4	Update whit fixes	15/05/26	LM	MD	MPV	CLIENT: MMA			62	6	
3	For Manufacturing	12/03/26	LM	MD	MPV						
REV	DESCRIPTION	DATE	PROJ.	DW.	APR.			CONT: 81			

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NOTE: cables not indicated will be black 0.75mm²/18 AWG

NOTA

1) El switch SW9 en la placa electronica E2.5 habilita o deshabilita la resistencia de terminación de 120Ω en la linea RS-485 del PORT #2.

-> SW9 ON: Resistencia de terminación activada (120Ω entre A y B)

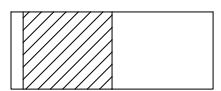
-> SW9 1: Resistencia de terminación desactivada.

NOTA

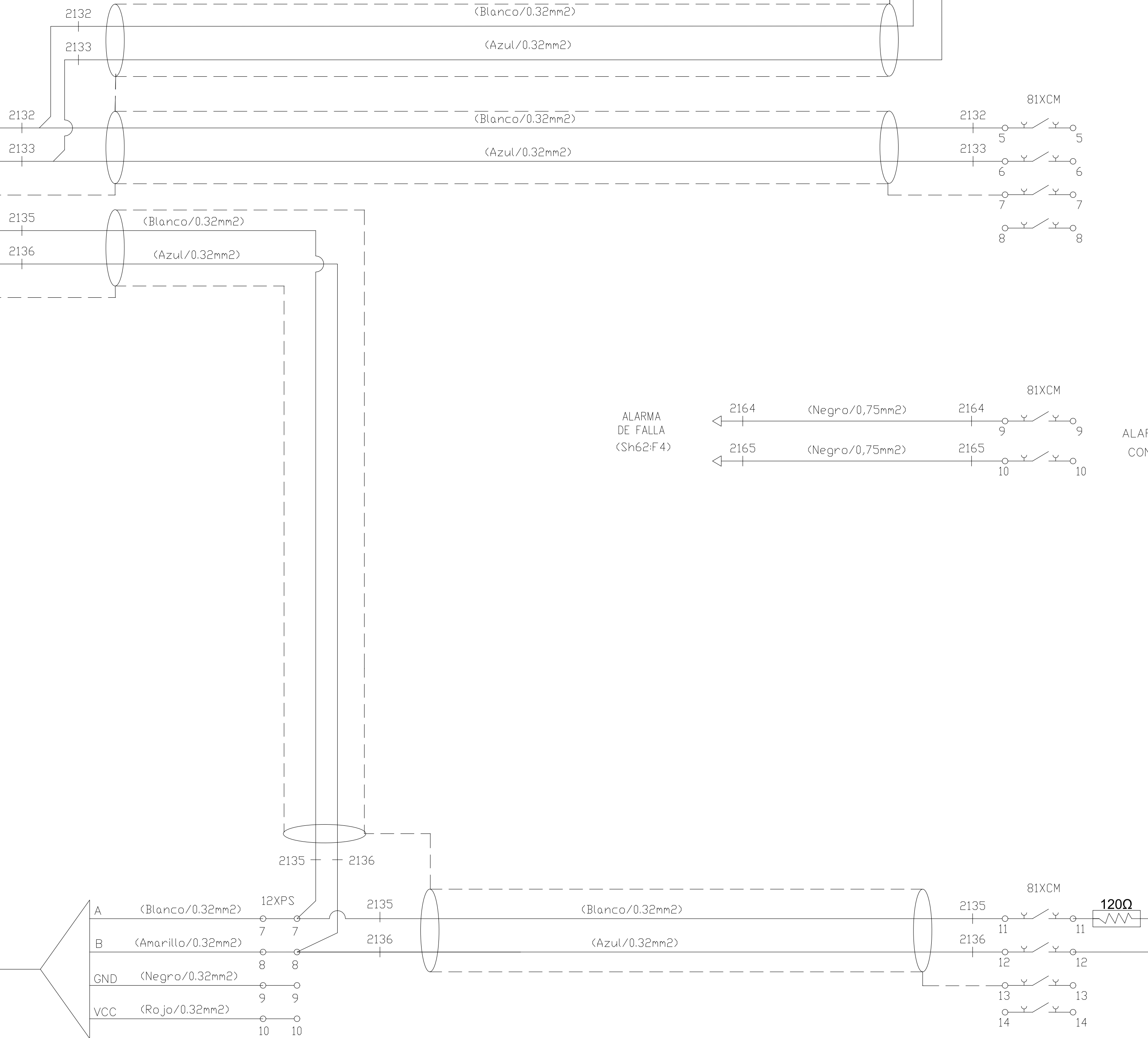
Los cables de comunicación RS 485 son marca Marlew, de 1x2x22 AWG (0,32 mm²)
En los terminales de conexión de cada equipo se admite una distancia máxima de 25mm del par desenrollado y sin malla

UNIDAD DE CONTROL E2.5
(PRINCIPAL)

21PCB01

SW9 ON  1

Communication Port #2	RS485 A	32
	RS485 B	33
	Return	34
Communication Port #1	RS485 A	35
	RS485 B	36
	Return	37



COMUNICACIÓN PARALELO
RS-485 (SCADA)
SALIDA

COMUNICACIÓN PARALELO
RS-485 (SCADA)
ENTRADA

SIN FUNCIÓN
RESERVA

NOTA

Sección máxima admisible para borne de conexión a campo
-Cable flexible con terminal:2,5mm² - 14 AWG.
-Cable rígido sin terminal:4mm² - 12 AWG.
-Cable flexible con dos conductores:0,5 mm² - 20 AWG.

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6	Update whit fixes	13/07/26	LM	MD	MPV
5	Update whit fixes	26/05/26	LM	MD	MPV
4	Update whit fixes	15/05/26	LM	MD	MPV
3	For Manufacturing	12/03/26	LM	MD	MPV
REV	DESCRIPTION	DATE	PROJ.	DW.	APR.

PROJECT: C.J. CONTROLADOR mSafe2

CODE: 75286-1

CLIENT:

MMA



DOC NUMBER: 75287

TITLE: DIAGRAMA ELÉCTRICO mSafe2
Comunicación

GRAL REV: 6

PAG: 81

REV: 6

CONT: